Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} S_g &= \int d^4x\, e \left\{ R(e,\omega) + 6g^2 + 2e^{-1}\epsilon^{\mu\nu\rho\sigma}\bar{\psi}_\mu\gamma_5\gamma_\nu \left(\hat{\mathcal{D}}_\rho + igA_\rho\sigma^2\right)\psi_\sigma - \mathcal{F}^2 \right. \\ & \left. + \mathcal{J}_{(m)}{}^{\mu\nu}(\mathcal{J}_{(e)\mu\nu} + \mathcal{J}_{(m)\mu\nu}) \right\} , \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} V_{12} &= 2^2 \cos[k_{1\lambda_1} \wedge k_{2\lambda_2} + (p_\lambda + q_\lambda) \wedge (k_{1\lambda_1} - k_{2\lambda_2})] \cos(p_\lambda \wedge q_\lambda), \\ V_{34} &= 2^2 \cos[k_{3\lambda_3} \wedge k_{4\lambda_4} + (p_\lambda + q_\lambda) \wedge (k_{3\lambda_3} - k_{4\lambda_4})] \cos(p_\lambda \wedge q_\lambda), \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{I}\left(\lambda\delta + (1-\lambda)\delta'\right) &\geq \lambda \sup_{\gamma\in\Sigma(\delta)} \int_{\mathcal{S}} f(s)d\gamma(s) + (1-\lambda) \sup_{\gamma'\in\Sigma(\delta')} \int_{\mathcal{S}} f(s)d\gamma'(s) \\ &\geq \lambda\mathcal{I}(\delta) + (1-\lambda)\mathcal{I}(\delta'). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{gathered} [e_i, f_j] = \delta_{ij}\tfrac{k_i - k_i^{-1}}{Q_i - Q_i^{-1}} \\ k_i e_j k_i^{-1} = Q_i^{a_{ij}} e_j, \quad k_i f_j k_i^{-1} = Q_i^{-a_{ij}} f_j \\ k_i k_i^{-1} = k_i^{-1} k_i = 1, \quad k_i k_j = k_j k_i \\ [e_i, e_i] = [f_i, f_i] = 0, \text{if} \quad a_{ii} = 0 \end{gathered}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{R}_{\beta_0,\beta_1}(z) :=& (\mathcal{A}_0 - zI)^{-1} - \left(I - z\mathcal{A}_0^{-1}\right)^{-1} \Pi(\overline{\beta_0 + \beta_1 M(z)})^{-1}\beta_1\left(\Pi^* \left(I - z\mathcal{A}_0^{-1}\right)^{-1}\right) \\ beginalign*0.3em] =& (\mathcal{A}_0 - zI)^{-1} - S(z)(\overline{\beta_0 + \beta_1 M(z)})^{-1}\beta_1 S^*(\overline{z}). \end{aligned}$$